
Warm-up Characterization and an Open-Weight Harness for Lean Refactor Arena

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Abstract

1 The Lean Refactor Arena at the NeurIPS 2026 VeriCodeGen workshop asks
2 whether agents can make Lean 4 proofs better, not merely correct, by reducing
3 proof size, reducing elaboration cost, and preserving compilation across toolchains.
4 We import the public warm-up JSONL (15 theorems from Strata, PhysLib, CSLib,
5 ArkLib, and PutnamBench) and report a frozen characterization of that subset. We
6 then describe an open-weight refactoring harness that uses a local Leanstrat owner
7 to propose tactic-level rewrites while Lean remains the checker. This report does
8 *not* claim Arena leaderboard scores, token savings, or a completed official submission:
9 the warm-up characterization is measured; the refactoring run is specified
10 and unrun.

11 1 Approach

12 The competition is a two-track verified proof-refactoring task [8, 5, 4]. Track 1 wraps closed-source
13 APIs under a US\$3-per-problem cap. Track 2 post-trains or harnesses open-weight models and
14 must run the full benchmark on at most four 80 GB A100s in 48 hours. Scoring has three axes:
15 proof-source token count, Lean elaboration effort, and zero-shot version transfer. Theorem statements
16 must be preserved and proofs must remain Lean-correct.

17 This report targets Track 2 as a *development* harness on a local NVIDIA GB10 DGX Spark owner,
18 not as an official $4\times$ A100 run. The pipeline is:

- 19 1. Load a problem from `benchmark_data_warmup.jsonl`: theorem name, statement, refer-
20 ence proof, file path, and toolchain snapshots.
- 21 2. Construct a prompt that includes the statement, the reference proof, and a request for a
22 shorter tactic script that proves the same statement.
- 23 3. Call a local OpenAI-compatible Leanstrat server (`leanstrat_local` on
24 `172.17.0.1:8080`) with temperature 0.
- 25 4. Replace the proof body, compile with the listed Lean snapshots, and record token count and
26 elaboration metrics only if compilation succeeds.
- 27 5. Reject any candidate that changes the statement, fails to compile, or depends on untrusted
28 oracles.

29 Lean is the only correctness oracle. Failed candidates are retained as measured failures, not rewritten
30 as successes. We have not yet executed this loop on the 15 warm-up problems; Section 1.1 reports
31 only the imported reference proofs.

32 **Intended prompt (unrun).** The generator sees the theorem `statement`, the reference `src`, and
33 the instruction: return only a tactic proof of the same statement that is shorter in source tokens; do
34 not change the statement; do not invent axioms. The checker then substitutes the candidate body and
35 compiles. No LLM-as-judge step exists.

Validity and scoring. A candidate is valid only if (i) the declaration is byte-identical to the JSONL statement (modulo whitespace in the proof body), (ii) it compiles on every Lean tag in that record’s `version_info`, and (iii) no extra axiom or sorry is introduced. Invalid candidates score as failures on all three organizer axes: proof-source token count, elaboration effort, and zero-shot version transfer [8]. We will report those axes only after Lean receipts exist. Aggregating a “win” from a subset of snapshots, or from Spark-only compiles, is out of scope.

1.1 Warm-up subset

We downloaded the public warm-up JSONL from the Arena Gradio API on 2026-09-16. The file contains 15 records (113 826 bytes), three problems from each of Strata [7], PhysLib [6], CSLib [2], ArkLib [1], and PutnamBench. SHA-256 of the frozen copy is 6209680cf00cde0765b77b24834cd72c64dd585b2f7e3f2a58209980ab59a804. The Gradio `/tmp/gradio/` URL is ephemeral; the local copy is authoritative.

Table 1: Public warm-up subset (15 problems). Proof tokens are the JSONL `proof_length` field. Short names are last-component labels; full identifiers are in the appendix. No refactoring scores are reported.

Source	Theorem (short)	Lines	Tokens	Toolchains
Strata	<code>substOldPostSubset</code>	97	313	1
Strata	<code>extractedOldExprInVars</code>	82	222	1
Strata	<code>InitsUpdatesComm</code>	55	224	3
PhysLib	<code>var. calculus FT</code>	126	851	1
PhysLib	<code>time-deriv E extrema</code>	126	1372	1
PhysLib	<code>iota time-order</code>	117	1217	1
CSLib	<code>Fsub.Typing.progress</code>	83	408	3
CSLib	<code>parallelReduction diamond</code>	64	407	4
CSLib	<code>bisimilarity congr choice</code>	62	251	3
ArkLib	<code>fiberwise unique-dec.</code>	209	1463	4
ArkLib	<code>fold advances eval poly</code>	184	1906	3
ArkLib	<code>interleaved affine gaps</code>	177	1353	3
Putnam	<code>putnam_1964_a4</code>	186	1769	3
Putnam	<code>putnam_1964_b2</code>	294	1754	2
Putnam	<code>putnam_1995_a3</code>	282	4085	3

Reference proof length ranges from 222 to 4085 tokens (mean 1173.0). Line counts range from 55 to 294 (mean 142.9). Source-file character counts in the `src` field range from 1727 to 13 801 (mean 6183.0). Putnam problems ship without a `file_path/url` and are the only records with a nonempty header (Mathlib/Aesop imports); the others point at public Lean repositories. Toolchain multiplicity varies: some theorems list a single Lean snapshot, others list three or four (v4.25.0–v4.33.0-rc2). That spread is the version-transfer axis of the official scorer.

JSONL schema. Each record has the twelve fields in Table 2. The `version_info` field is a list of single-key maps from Lean tag to git commit. We do not treat `proof_length` as an official Arena score; it is the organizer-supplied size of the reference proof.

Per-source notes. Strata problems are CallElim/Core lemmas on Lean v4.26–v4.29, short (222–313 tokens). PhysLib problems are variational calculus, electromagnetism extrema, and Wick-algebra time-order lemmas, all pinned to v4.32.0 only, so they cannot yet exercise version transfer. CSLib problems (Fsub progress, SKI diamond, CCS bisimilarity) list three or four snapshots including v4.33.0-rc2. ArkLib problems are the longest non-Putnam proofs (1353–1906 tokens) with v4.28–v4.31 snapshots. Putnam proofs are the longest overall (1754–4085 tokens) and the only records missing repository URLs.

Competition phase. The workshop timeline lists warm-up as now through 30 September 2026, with the full benchmark on 1 November 2026 and proof/code plus tech-report deadline on 8 November 2026 [8]. This report is a warm-up census during that first phase. It is not a full-benchmark entry.

Table 2: Warm-up JSONL fields. All 15 records share this schema.

Field	Role
name	Fully qualified theorem identifier
source	strata/physlib/cslib/arklib/putnambench
statement	Declaration to preserve
src	Reference proof source
proof_length	Organizer reference token count
num_lines	Reference line count
header	Imports; nonempty only for Putnam
file_path, url	Repo path; empty for Putnam
start_line, end_line	Span in the upstream file
version_info	Lean tag $\mapsto$ commit snapshots

2 Models

The intended generator is **Leanstral-1.5-119B-A6B-NVFP4** [3], served locally as `leanstral_local` by `llama-server` bound to the Docker bridge (172.17.0.1:8080). Weights are the NVFP4 GGUF already used as the VeriCodeGen generated-code owner on this host. Sampling for the unrun harness is temperature 0, seed equal to the problem index, maximum 1024 completion tokens, and a 2048-token input ceiling. No embedding model, reranker, or LLM judge is in the pipeline. Lean snapshots listed in each JSONL `version_info` entry are the only compilers. We have not fine-tuned Leanstral for this competition.

This host is a DGX Spark with one GB10 GPU, not the official Track 2 4×A100 80 GB envelope. Development measurements on Spark must not be reported as official open-track scores.

3 Budget accounting

This report Importing and characterizing the 15-line JSONL used CPU only (seconds of wall-clock; \$0 API). No closed-track API spend occurred. No Track 1 or Track 2 inference run was billed.

Intended Track 2 run Official scoring must use $\leq 4 \times A100$ 80 GB and ≤ 48 hours for the full benchmark. Fine-tuning compute, if any, will be reported separately and is not capped. We have not started that run. The hardware table (Table 3) records a *planned* envelope, not a measured one.

Table 3: Open track: planned hardware. The benchmark-run row is unrun. Only an official 4×A100 inference run would count against the 48-hour cap.

Stage	Accelerators	Wall-clock
Warm-up characterization (this report)	CPU	<1 min (measured)
Development harness (Spark GB10)	1 × GB10	unrun
Official Track 2 benchmark run	4 × A100 80 GB	unrun; cap 48 h

We do not claim general token savings. Negative and incomplete outcomes will be retained when the harness runs. Under-reporting cost, or reporting Spark wall-clock as if it were the official envelope, would be a desk-rejection condition [8]; we therefore leave the official row blank rather than fill it with a substitute.

4 Reproduction

The imported JSONL and the machine-readable summary live at `papers/completion/lean_refactor_arena/data/`. Regenerating Table 1 and the SHA-256 check is

```
python3 papers/completion/lean_refactor_arena/tools/summarize_warmup.py
```

93 from the repository root. The command fails closed if the JSONL digest is not
 94 6209680cf00cde0765b77b24834cd72c64dd585b2f7e3f2a58209980ab59a804. The Arena
 95 UI is <https://huggingface.co/spaces/delta-lab-ai/lean-refactor-arena>. Work-
 96 shop rules: <https://vericodegen.github.io/>. Competition site: <https://leanrefactor.github.io/>.

98 The refactoring harness is specified above and is **not** executed in this report. Organizers cannot
 99 regenerate a submission JSONL from this archive yet: there is no entry-point that emits refactored
 100 proofs. When that entry point exists, the README will give the command, expected wall-clock, and
 101 hardware. Sampling at temperature 0 is intended to be deterministic given the local owner; we have
 102 not measured score variance across reruns because we have not run the scorer.

103 Desk-rejection conditions we are already satisfying or leaving blank rather than faking: the four
 104 required section headings are present; every model that would be used is named; Track 1 spend is \$0;
 105 Track 2 wall-clock is unrun rather than substituted; no auxiliary judge is hidden.

106 5 Limitations

107 This is a competition-report scaffold plus a frozen warm-up census. It is not a ranked Arena entry.
 108 We have not compiled the 15 proofs, have not measured elaboration, and have not submitted to the
 109 Space. Putnam records lack repository URLs in the JSONL. Local Spark hardware is not the official
 110 Track 2 envelope. The three scoring axes remain unspecified numerically for our system because no
 111 candidate proofs exist.

112 Acknowledgments and Disclosure of Funding

113 Workshop: VeriCodeGen, NeurIPS 2026. Competition: Lean Refactor Arena (delta-lab-ai). No
 114 Arena submission was made with this report.

115 References

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- 125 [7] Strata. <https://github.com/strata-org/Strata>, 2026.
- 126 [8] VeriCodeGen: AI for verifiable coding — NeurIPS 2026 workshop. <https://vericodegen.github.io/>,
 127 2026. Accessed 2026-09-16.

128 A Full warm-up identifiers

129 The short-name table (Table 1) uses abbreviated labels. The frozen JSONL names, repositories, and
 130 Lean tags follow.

- 131 • `CallElimCorrect.substOldPostSubset` — Strata,
 132 Strata/Transform/CallElimCorrect.lean, v4.26.0, 97 lines, 313 tokens.
- 133 • `CallElimCorrect.extractedOldExprInVars` — Strata, same file, v4.26.0, 82 lines,
 134 222 tokens.

- 135 • `Core.InitsUpdatesComm` — Strata, `Strata/Languages/Core/`
136 `StatementSemanticsProps.lean`, v4.29.1/v4.27.0/v4.26.0, 55 lines, 224 tokens.
- 137 • `fundamental_theorem_of_variational_calculus'` — PhysLib,
138 `Physlib/Mathematics/VariationalCalculus/Basic.lean`, v4.32.0, 126 lines, 851
139 tokens.
- 140 • `Electromagnetism.ElectromagneticPotential.`
141 `time_deriv_time_deriv_electricField_of_isExtrema` — PhysLib,
142 `Physlib/Electromagnetism/Dynamics/IsExtrema.lean`, v4.32.0, 126 lines, 1372
143 tokens.
- 144 • Wick-algebra time-order (JSONL name uses Greek iota) — PhysLib, `Physlib/QFT/`
145 `PerturbationTheory/WickAlgebra/TimeOrder.lean`, v4.32.0, 117 lines, 1217
146 tokens.
- 147 • `Cslib.LambdaCalculus.LocallyNameless.Fsub.Typing.progress` — CSLib,
148 `Cslib/Languages/LambdaCalculus/LocallyNameless/Fsub/Safety.lean`,
149 v4.33.0-rc2/v4.32.0/v4.31.0, 83 lines, 408 tokens.
- 150 • `Cslib.SKI.parallelReduction_diamond` — CSLib, `Cslib/Languages/`
151 `CombinatoryLogic/Confluence.lean`, four snapshots through v4.30.0, 64 lines, 407
152 tokens.
- 153 • `Cslib.CCS.bisimilarity_congr_choice` — CSLib, `Cslib/Languages/CCS/`
154 `BehaviouralTheory.lean`, three snapshots, 62 lines, 251 tokens.
- 155 • `Binius.BinaryBasefold.`
156 `fiberwise_dist_lt_imp_dist_lt_unique_decoding_radius` — ArkLib,
157 `ArkLib/ProofSystem/Binius/BinaryBasefold/Prelude.lean`, four snapshots
158 v4.28.0–v4.31.0, 209 lines, 1463 tokens.
- 159 • `Binius.BinaryBasefold.fold_advances_evaluation_poly` — ArkLib, same file,
160 three snapshots, 184 lines, 1906 tokens.
- 161 • `interleaved_affine_gaps_imply_tensor_gaps` — ArkLib, `ArkLib/Data/`
162 `CodingTheory/ProximityGap/DG25/MainResults.lean`, three snapshots, 177 lines,
163 1353 tokens.
- 164 • `putnam_1964_a4`, `putnam_1964_b2`, `putnam_1995_a3` — PutnamBench; empty
165 `file_path/url`; Mathlib+Aesop headers; 186/294/282 lines and 1769/1754/4085 tokens.